

ACUTE ISCHAEMIC STROKE AND **TRANSIENT ISCHAEMIC ATTACK**

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Updated in April 2024 by Dr Tan Wei Beng. Next review date: April 2026.

CONTENTS

Introduction	3
Importance	3
Epidemiology	3
Definition	3
Assessment	4
Symptoms and signs	4
Differential Diagnoses	5
Initial Assessment of Stroke / TIA in Primary Care:	6
Investigations	8
Management and Follow-Up	9
Post-Stroke / TIA Step-Down Management:	9
Recommended Follow-Up Interval	10
Specific Points on Anti-Platelets / Warfarin and Direct Oral Anticoagulant Use	10
Surgical Intervention	11
Role of Health Team Members	12
Referral to Specialist	15
When to Refer to Specialist	15
Special Situations	15
Fitness to Drive	15
Community Resources	16
Recommended Care Components for Stroke	17
Abbreviations	18
References	19

I. INTRODUCTION

Importance

Cerebro-vascular disease accounts for more than 10,000 admissions in Singapore annually and is a leading cause of adult neurological disability. About 25% recover with minor impairment, and 40% experience moderate to severe impairments requiring special care. The recurrence rate is 5 to 15 % per year.

Transient ischaemic attacks (TIA) are harbingers of stroke with subsequent risk of stroke being around 10% in the first 90 days after a TIA.

For any acute onset focal neurological deficit, it remains a challenge to differentiate between acute stroke / TIA and other stroke mimics, hence rapid initial evaluation and early neuro-imaging is vital to achieve higher diagnostic certainty and facilitate timely intervention to improve functional outcomes of patients suffering from stroke.

NUP receive post-stroke patients discharged from the hospitals for follow-up care. Readmissions to hospital can be due to recurrent stroke or post-stroke complications. This can be reduced with appropriate screening to detect post-stroke complications and early intervention.

Epidemiology

Stroke was the fourth most common cause of death in 2020, accounting for 6.0% of all deaths in Singapore.

Based on data published by the Singapore Stroke Registry, there is a significant increase in the crude incidence rate of strokes between 2010 and 2020, with 5,890 episodes in 2010 and 8,846 episodes in 2020 respectively. About 80% of stroke incidence each year were ischaemic strokes (IS), while 20% were haemorrhagic strokes (HS).²

Definition

Acute ischaemic stroke (AIS) is defined as “An episode of neurological dysfunction caused by focal cerebral, spinal, or retinal infarction”³ whereas infarction is further defined as brain, spinal cord, or retinal cell death attributable to ischemia, based on

1. Pathological, imaging, or other objective evidence of cerebral, spinal cord, or retinal focal ischemic injury in a defined vascular distribution; or
2. Clinical evidence of cerebral, spinal cord, or retinal focal ischemic injury based on symptoms persisting ≥ 24 hours or until death, and other aetiologies excluded.

Focal arterial ischemia with transient symptoms (lasting < 24 hours) and without evidence of infarction by pathology or imaging should be considered a TIA.

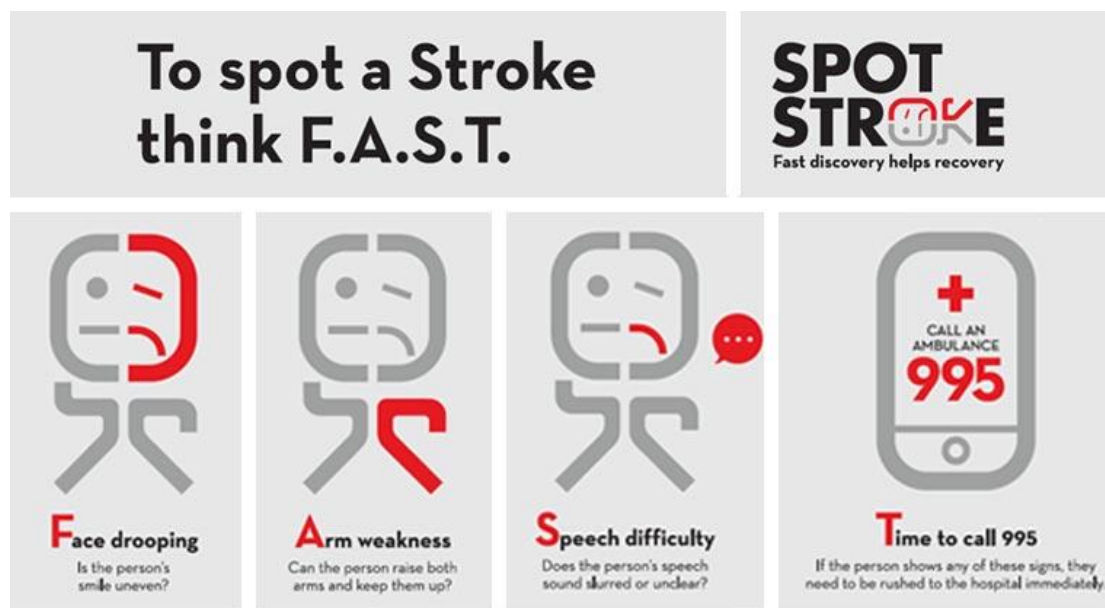
II. ASSESSMENT

Symptoms and signs

Symptoms of acute ischaemic stroke and TIA are typically sudden in onset and can include:

1. Facial asymmetry⁺
2. Arm and leg weakness⁺, especially on one side of body
3. Slurring of speech⁺ (dysarthria)
4. Loss of vision in one eye or one hemifield (homonymous hemi- or quadranopia)
5. Double vision (diplopia)
6. Unsteadiness of gait, loss of balance or coordination
7. Difficulty in understanding or expression of language (dysphasia)
8. Vertigo in the presence of any of the above symptoms

⁺1, 2 and 3 are part of the acronym FAST (Face Arm Speech Time), used widely in public health messaging on the symptoms of TIA / stroke and the emphasis to seek immediate medical attention upon onset of symptoms. Patients with cardiovascular risk factors should be educated on FAST. The following illustration serves as a quick guide in recognising stroke symptoms.



NB: Patients presenting with giddiness alone or numbness alone should be separately evaluated due to the wider range of differentials. Primary care physicians are encouraged to familiarize and adopt **ABCD² Score for TIA** (available on MDCalc website) when evaluating patients presenting with transient neurological symptoms suspicious for TIA. Low ABCD² score (0-3 points) represent low stroke risk at 7-day and 90-day.

Differential Diagnoses

The important differential diagnoses include:

A) Persistent neurologic deficits with abrupt onset

- Intracerebral haemorrhages
- Brain tumours
- Brain abscesses
- Non-ketotic hyperglycaemic stupor
- Attacks of multiple sclerosis
- Demyelinating lesion in brain - acute disseminated encephalomyelitis (ADEM)

B) Transient neurologic events

- Seizure / post-ictal paralysis
- Complicated migraine/ hemiplegic migraine/ migraine aura
- Syncope
- Metabolic perturbations, such as hypoglycaemia
- Hepatic, renal, and pulmonary encephalopathies
- Pressure- or position-related peripheral nerve or nerve root compression – can cause transient numbness
- Peripheral vestibulopathies – can cause transient episodic dizziness / vertigo

Initial Assessment of Stroke / TIA in Primary Care:

Sudden onset or noted on awakening:

- Facial asymmetry
- Arm and leg weakness and / or numbness
- Slurring of speech
- Unsteadiness of gait, loss of balance/ coordination
- Loss of vision in one eye or one hemifield, double vision
- Difficulty in understanding or expression of language
- Vertigo associated with any of the previously mentioned symptoms



1. Check blood glucose for hypo / hyperglycaemia
 2. Check ECG for atrial fibrillation or myocardial infarction
- *Empiric loading of anti-platelets without brain imaging while awaiting transfer to acute hospital is not recommended in view of possibility of haemorrhagic infarct



Urgent referral

- Early brain imaging from symptoms onset is recommended.
- Referral to Emergency Department (ED) should be considered for all patients with suspected TIA or acute ischaemic strokes.
- Activation of the 995 system is **STRONGLY** recommended in patients with suspected stroke to minimise delay in transportation to hospital for potential interventions in eligible patients*⁴

*IV thrombolysis with alteplase (a recombinant tissue plasminogen activator a.k.a. rtPA) is recommended for selected patients who can be treated within 4.5 hours of ischemic stroke symptom onset or patient last known well or at baseline state.

*Patients are eligible for mechanical thrombectomy if they meet all the following criteria: (1) pre-stroke mRS score of 0 to 1; (2) causative occlusion of the internal carotid artery or middle cerebral artery (MCA) segment 1 (M1); (3) age ≥18 years; (4) NIHSS score of ≥ 6; (5) ASPECTS of ≥ 6; and (6) treatment can be initiated (groin puncture) within 6 hours of symptom onset.

*In selected patients with AIS within 6 to 24 hours of last known normal who have Large Vessel Occlusion (LVO) in the anterior circulation and meet other DAWN or DEFUSE 3 eligibility criteria, mechanical thrombectomy is reasonable.

*In selected patients with AIS who have LVO in the posterior circulation, mechanical thrombectomy is reasonable based on emerging evidence.

Where:

- mRS: Modified Rankin Scale
- NIHSS: National Institute of Health Stroke Scale
- ASPECTS: Alberta stroke programme early CT score

**Centres with thrombolysis and thrombectomy capabilities include: NUH, SGH, TTSH*

Investigations

Investigations are performed to rule out the differential diagnoses, detect risk factors and determine the stroke mechanism, so that appropriate therapy may be offered. However, performing these investigations should not delay the activation of 995 system and/or transfer of patient to emergency department.

Initial investigations which can be performed in primary care setting (provided no delay in transfer of patient via ambulance) include:

- Blood glucose, POCT
- Electrocardiogram (ECG)
- Full blood count, with platelet count (FBC)
- Renal function with creatinine (RP)
- Coagulation profile (PT / INR / aPTT)

At hospitals

- Echocardiogram and prolonged cardiac rhythm monitoring
- Brain imaging – CT or MRI Brain
- Angiographic studies – ultrasound (carotid duplex, transcranial Doppler), CT or MR angiography

III. MANAGEMENT AND FOLLOW-UP

Post-Stroke / TIA Step-Down Management:

Majority of the stroke and TIA patients are discharged or handed over to primary care physicians for long term management after recovering from the initial period of stroke / TIA.

The following actions are recommended for management in the primary care setting:

1. Ensure medication are optimised for the patient's profile [Anti-thrombotics i.e., antiplatelet (with specified choice and duration) and anticoagulation as specified by the hospital, lipid lowering agents, anti-hypertensive medications].
2. Ensure monitoring of lipid profile, fasting glucose / HbA1c, GFR / ACR and any other test as clinically indicated (for instance, INR for warfarin and FBC for anticoagulation therapy) at interval deemed necessary by the primary care physician and appropriate follow up action on the test results.
3. Manages the modifiable risk factors accordingly as stated below.
4. Advice for flu and pneumococcal vaccination as recommended under the National Adult Immunisation Schedule
5. Assess baseline rehabilitation need for Activities for Daily Living (ADL), i.e., mobility, feeding, dressing, toileting, washing or bathing, transferring.
6. Assess suitable patient for any frailty or any need of further support from the community, refer to [Section V](#)

The modifiable risk factors of stroke and TIA which should be addressed include:

1. **Hypertension:** Avoid aggressive treatment of elevated blood pressure in the initial few days after an acute stroke or TIA. In the follow-up phase, BP should be maintained within normotensive range i.e. below 140/90 mmHg, or below 130/80 mmHg in the presence of concomitant cardiovascular risk factors. Refer to CPG on [Hypertension](#)
2. **Hyperlipidaemia:** The target LDL-c should be set at a lower value (< 1.8 mmol/L) for patients with history of stroke and TIA, consider moderate to high intensity lipid lowering agents. Refer to [ACE's Lipid management guideline](#)
3. **Diabetes mellitus:** Maintain HbA1c below 7%. Refer to CPG on [Diabetes Mellitus](#)
4. **Atrial fibrillation:** Discuss anticoagulation (see below)
5. **Heart disease:** Optimise treatment of coronary artery disease, heart failure, arrhythmias and valvular heart disease
6. **Smoking:** Smoking cessation counselling / clinic
7. **Excessive alcohol usage:** To take alcohol in moderation.
8. **Sedentary lifestyle:** Exercise advice / prescription
9. **Obesity:** Weight management advice / programme

Recommended Follow-Up Interval

Criteria	Recommended TCU Frequency	Recommended for alternate Dr / CM visit
- Good blood pressure, AND - Elderly or has complications such as IHD, CVA, renal impairment	3 - 4 months [^]	√
- Adherent to follow-up and treatment, AND - With or without co-morbidities / complications, AND - Stable but sub-optimal control related to individual targets for BP, HbA1C and Cholesterol over the past 4 -6 months	3 - 4 months	√
- Poor blood pressure control, AND - Requires titration of medication	2 weeks	

[^] 6 months is acceptable too

Specific Points on Anti-Platelets / Warfarin and Direct Oral Anticoagulant Use

1. Antiplatelet therapy reduces the risk of recurrent stroke, myocardial infarction and vascular death by 23%.
Short course DAPT followed by long-term SAPT is usually initiated at time of discharge from hospital to reduce the risk of early recurrent strokes. Beyond 90 days after stroke, DAPT is associated with increased risk of bleeding and no benefit in long-term reduction of recurrent stroke risk with exception of certain indications such as stent placement, or medical indications due to conditions from other system such as cardiac or vascular surgery. Hence, the choice of DAPT combination and its duration is often initiated by the hospital stroke physicians at time of discharge, and the expected date of transition to long term SAPT is essential during medication review by the primary care physicians / pharmacists.
 - a) Aspirin - usual dose 100 mg/day
- caution: gastrointestinal bleeding especially with higher doses of aspirin
 - b) Clopidogrel - 75 mg/day for patients intolerant to Aspirin or who have peptic ulcer disease
 - c) Ticagrelor - 90 mg BD, currently not in NUP formulary, caution in patients with history of gout/hyperuricemia
 - d) Dipyridamole - 225-400 mg/day headache / flushing with dipyridamole, hence titrate dose upward slowly
 - e) Ticlopidine - 250 mg bid-caution: neutropenia; check FBC two weekly for the first 3 months

(Ticagrelor has not been incorporated into guideline for monotherapy in secondary stroke prevention at present.
Dipyridamole is reserved for patient who is unable to tolerate clopidogrel and rarely initiated in the NUHS cluster hospital setting at present.
Ticlopidine is no longer used in the NUHS cluster hospital setting)
2. Anticoagulation therapy reduces the risk of recurrent stroke following a TIA or a minor stroke in patients with atrial fibrillation by approximately two-thirds.

- a) Patients with atrial fibrillation or valvular heart disease should be evaluated by a cardiologist, if not evaluated previously.
- b) Patients with atrial fibrillation should be counselled on the benefits and risks associated with anticoagulation.
- c) The option of newer oral anticoagulants such as DOAC could be discussed with the patient.
- d) Anticoagulation therapy is recommended for those with cardioembolic stroke / TIA.
- e) For DOAC prescription, ensure appropriate dosing according to the patient's renal status in terms of creatinine clearance (for rivaroxaban), age, body weight and creatinine level (for apixaban) (please refer to [NUP CPG - Atrial Fibrillation](#) for DOAC).
- f) For patients treated with warfarin, the target International normalised ratio (INR) is usually 2.0-3.0 but may be individualised based on patient profile the treating cardiologist. If in doubt, to check with hospital ACC team before continuation.

Surgical Intervention

Some of suitable stroke / TIA patients may be offer further surgical interventions such as:

1. Carotid endarterectomy or carotid artery stenting can be considered in patients with ipsilateral internal carotid stenosis of 70 - 99% causing symptomatic stenosis, where the risk of stroke after TIA is high.
2. Closure of patent foramen ovale for eligible patients whose stroke ethology is due to paroxysmal right-to-left shunt

Decision for surgical intervention should be deferred to stroke neurologist via referral to assess indication and patient eligibility.

Role of Health Team Members

1. Family Physician

- Able to provide all aspects of primary medical care from screening, diagnosis and management of dementia, including health promotion and prevention / treatment of complications.
- Collaborates with other health care providers in the health team to provide holistic and patient-centred care.

2. Care Coordinator

- Introduce OneNUHS and HealthHub Apps to patient.
- Introduce and enrol patient onto Teleconsult.
- Perform PHQ2, frailty and functional screenings, abbreviated mental test (AMT) and comprehensive geriatric assessment (CGA) when indicated.
- Address care gaps under Health Maintenance Topics, such as vaccinations (influenza / pneumococcal), cancer screenings (FIT kit, mammogram, PAP smear / HPV).

3. Care Manager

- Provide basic education and materials on AIS/TIA for patients and caregivers Provide basic education and materials on AIS/TIA for patients and caregivers, including health promotion and prevention.
- Assess for treatment adherence and liaise with other healthcare personnel to close care gaps.

4. Advanced Practice Nurse

- Co-manage patients who are diagnosed with mild and stable AIS/TIA (e.g., symptoms review, post-medication review) including pre-existing chronic conditions with physician and health team.
- Liaise with other healthcare personnel to close care gaps and provide holistic and patient-centred care.

5. Clinical Pharmacist

- Manage patients with diabetes, hypertension and dyslipidaemia.
- Initiate and titrate medications for secondary stroke prevention.

6. Financial Counsellor

- Financial counselling for patients with issues on payment of medical bills.
- Advise on MediSave utilisation for chronic disease medical bills.
- Application on government endowment fund- MediFund and Medication Assistance Fund for subsidies on high-cost drugs.
- Application on Financial Assistance Scheme for eligible non-residents.

7. Dietitian

- Dietary management for patients with malnutrition or poor appetite / oral intake.
- Dietary management for patients with swallowing difficulties/enteral feeding.
- Dietary management for patients with chronic diseases i.e., hypertension, diabetes, hyperlipidaemia.
- Dietary management for patients on warfarin.

8. Medical Social Worker

- Conduct biopsychosocial assessment.
- Provide intervention for care-giver stress, social assistance including application of disability assistance schemes (e.g., Seniors' Mobility and Enabling Fund (SMF), Pioneer Generation Disability Assistance Scheme (PioneerDAS), Migrant Domestic Worker Levy Concession, Severe Disability Assessment for ElderShield or CareShield Life), and referral for community resource / services and caregiver support (day-care, day-rehab or nursing home).
- Application of Enhanced for Active Seniors (EASE) for subsidised home modifications, and Senior's Mobility and Enabling Fund (SMF) for mobility and assistive devices and home healthcare items (e.g. catheters, milk supplements, thickeners, adult diapers, nasal tubing and wound dressings where eligible).
- Consider alternative financial support (not related to medical bills) when necessary.
- Referrals to AIS/TIA related care in the community when needed.

9. Psychologist

- Emotional and psychological support to navigate post-stroke adjustment and challenges.
- Psychological assessment and intervention to manage mood challenges (depression, anxiety *etc.*), cognitive impairment, and behavioural issues associated with AIS/TIA.
- Support patient and caregiver in developing strategies to improve daily functioning and quality of life.
- Psychological support for carers experiencing caregiver stress.

10. Physiotherapist

- Patients undergoing acute/sub-acute stroke rehabilitation should be referred and managed at a facility with rehab equipment and higher frequencies of rehab e.g., Day Rehab Centres, Community Hospitals.
- Patient education regarding physical activity and reducing sedentary periods.
- Walking aids prescription and gait training.
- Exercise prescription on resistance, flexibility, balance, and aerobic exercises to improve physical function.

11. Podiatrist

- To manage painful pathological callus and nails.
- Footwear and foot care education.

12. Pharmacist

- Smoking cessation clinic.
- Detect, prevent and address drug-drug / drug-disease interactions.
- Perform medication reconciliation.

IV. REFERRAL TO SPECIALIST

When to Refer to Specialist

1. New diagnosis of AIS/TIA – refer to hospital A&E.
2. Patients suspected of new stroke – refer to hospital A&E.
3. Patients with cognitive impairment or neuropsychiatric dysfunction.
4. Swallowing problems or oropharyngeal dysphagia – refer to rehabilitation medicine/ neurology for speech therapy

V. SPECIAL SITUATIONS

Fitness to Drive

After stroke / TIA, the patient's fitness to drive is assessed based on the Medical Guidelines on Fitness to Drive second edition 2011 by SMA as extracted as below:

For Group 1 - Class 1, 2 & 3 Licences

TIA: Single TIA – allowed to return to driving once free of TIA for 1 month.
Multiple TIA – allowed to return to driving once free of TIA for 6 months.

Stroke: Patients without any residual disability may resume driving after 1 month.
Patients with disabilities such as residual weakness, significant visual field defects, perceptual or mental impairment, in coordination, *etc.*, severe enough to interfere with control of the vehicle should not drive.
Patients with mild residual disabilities that may not interfere with control of the vehicle may undergo DARP* > 1 month after stroke. They may be allowed to drive if they pass DARP*, after final review by a neurologist.

For Group 2 - Class 4, 5 & Vocational Licences

TIA: Able to return to driving if all following conditions are met:

- > 6 months post-TIA (for single TIA) or > 1 year post-TIA (multiple TIAs or brainstem TIA)
- TIA is not due to high-risk underlying condition which is left untreated
- Compliant with treatment
- Final clearance given by a neurologist

Stroke: Able to return to driving if all the following conditions are met:

- 1-year post stroke
- Stroke is not due to high risk underlying condition which is left untreated (e.g. high grade carotid stenosis, untreated aneurysm)
- Fully recovered
- Passed DARP (Driving Assessment and Rehabilitation Programme)
- Compliant with treatment
- Final clearance given by a neurologist

VI. COMMUNITY RESOURCES

There are many free-of-charge or subsidised community resources available for post stroke patients, especially those who have disabilities. This list is not exhaustive but includes day rehabilitation service, day care service, home nursing service, home medical service, befriending services, meal deliveries, nursing home respite service, medical escort for medical appointment attendance and so on. For HDB-dwelling patients, there is home improvement program i.e., EASE program for installation of fall prevention measures such as non-slippery tiles/ grab bar / installation. There are also social and financial assistance available such as Pioneer Generation Disability Assistance Scheme (PGDAS), Home Caregiving Grant (HCG), Caregivers Training Grant (CTG), Senior Mobility and Enabling Fund (SMF). Certain subsidy and entitlements may only be eligible via means tested assessment.

It is advisable for primary care provider to conduct discussion with the patients and their family members before referring them to the medical social worker / financial counsellor for consideration to utilise these community resources to further enhance the community living of the stroke patients.

VII. RECOMMENDED CARE COMPONENTS FOR STROKE

Recommended Care Components	Minimum Frequency*	Remarks
Thromboembolism Risk Assessment[#]	As clinically indicated	Evaluate for atrial fibrillation, cardiac murmurs, fasting glucose and need for anti-thrombotic therapy
Rehabilitation Need Assessment	At baseline	
Blood pressure Measurement[#]	Twice a year	
Lipid Profile[#]	Annually	All patients should be risk stratified (as recommended in the Lipids CPG) Targets of treatment should be personalised by levels of risk
Smoking Assessment[#]	Annually for smokers; Once-off for non-smokers, unless there is a change in smoking habit	Assessment on smoking habits (estimated sticks/day; zero for non- or ex-smoker) and provide smoking cessation counselling
Influenza Vaccination[#]	Annually or per season	As recommended under the National Adult Immunisation Schedule (NAIS)

* More frequently if clinically indicated

[#] This is also a non-reportable clinical indicator

VIII. ABBREVIATIONS

ADEM	Acute disseminated encephalomyelitis
ASPECTS	Alberta stroke programme early CT score
AIS	Acute ischaemic stroke
DARP	Driving Assessment and Rehabilitation Programme
ECG	Electrocardiogram
ED	Emergency Department
FBC	Full blood count
HS	Haemorrhagic strokes
IS	Ischaemic strokes
LVO	Large vessel occlusion
MCA	middle cerebral artery
mRS	Modified Rankin scale
NAIS	National adult immunisation schedule
NIHSS	National Institute of Health Stroke Scale
rtPA	Recombinant tissue plasminogen activator
TIA	Transient ischaemic attacks

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